Programmable Sovereignty: Coordinated Specialization as a Response to the N-2 Problem in AI

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Abstract

The dominant scholarly and policy framing of artificial intelligence (AI) power treats sovereignty as a function of production: a state is sovereign over AI to the degree that it builds, owns, and can reproduce the technological stack on which AI depends. On this view, meaningful AI sovereignty is available only to the United States and, to a lesser extent, China — producing what we term the N-2 problem, in which every other state faces structural dependence on one or both. A separate and largely disconnected literature treats digital sovereignty as a function of governance: the capacity to shape how technology behaves through rules, standards, and market leverage, independent of who builds the underlying systems. This paper integrates these two literatures and, in doing so, makes two contributions. First, it defines programmable sovereignty precisely, as soft governance anchored by a deliberately minimal degree of hard-stack control — above all pooled compute capacity — sufficient to make governance commitments credible and dependence reversible, without requiring the costly and largely unattainable goal of full-stack AI autonomy. Second, the paper paper develops

coordinated specialisation as the institutional mechanism through which N-2 states can convert individual complementary capacities into a credible and minimally self-sufficient shared coalition AI stack. We illustrate the feasibility and limits of coordinated specialisation through three existing institutional precedents (EuroHPC, India Stack, and the ASEAN Digital Economy Framework) and close with policy implications for the European Union, Asia-Pacific coalition partners, and prospective coalitions generally.

Keywords: *AI governance, technology sovereignty, coordinated specialisation, industrial policy, digital sovereignty, US-China technology competition, regulatory leverage, compute governance*

Introduction

Artificial intelligence is commonly discussed as a race: a contest to build the largest models, accumulate the most patents, or announce the boldest national strategy, in which the central analytical task is to measure progress and project a winner. This framing, we argue, mischaracterises what is occurring. The evidence assembled in this paper indicates not a race with many credible entrants but a structural reorganisation of technological power in which only two states — the United States

and China — are positioned to approach full-stack control over the artificial intelligence production system, while every other economy faces a deepening and largely involuntary dependence on one or both.

Two recent episodes illustrate the mechanism by which this dependence is being converted into leverage. In April and October 2025, China extended export controls on rare earth elements to cover medium and heavy elements, associated processing technologies, and equipment, while introducing extraterritorial licensing requirements that apply to foreign-made products containing even trace quantities of Chinese-origin materials. In January 2026, Chinese authorities blocked Meta's proposed $2 billion acquisition of Manus, a Singapore-incorporated but China-origin artificial intelligence startup, and ordered the transaction unwound on national-security grounds related to the transfer of autonomous-agent technology and personnel (CNBC 2026). On 13 June 2026, the United States Department of Commerce gave Anthropic ninety minutes' notice to restrict access to two of its most capable models, Fable 5 and Mythos 5, to United States citizens only; Anthropic withdrew both models globally rather than implement the required access controls within the allotted time. These three episodes, occurring within an eighteen-month span and involving both members of the duopoly, are not isolated regulatory actions. They are demonstrations, in real time, of a structural condition that this paper terms the N-2 problem: in a world of N states, only two are capable

of pursuing full-stack technological sovereignty in artificial intelligence, and the remaining N-2 face a dependence that can be activated, at the discretion of either duopoly member, with little or no warning.

The N-2 problem extends an established line of analysis in international political economy concerning structural dependence under monetary hegemony — the costs to non-reserve-currency states of operating within a dollar-centered international monetary system (Eichengreen 2011; Cohen 2018; Helleiner 2014). We argue that AI dependence is materially more consequential than this N-1 currency analogy for three reasons developed in Section 2: it is more deeply infrastructural, penetrating the routine operation of public administration, healthcare, and security rather than only macroeconomic policy; the costs of switching away from an established AI stack are categorically higher than the costs of switching reserve currencies; and the window in which credible alternatives can still be constructed is closing as supply chains, governance norms, and data architectures are locked in.

Existing scholarship on AI power offers two only partially compatible responses to this problem, reviewed in Section 3. A first literature, which we term the hard-sovereignty tradition, treats sovereignty over AI as a function of productive capacity: ownership and control of the layers of the technology stack, from critical minerals and energy through to fabrication, foundation models, and capital (Lee 2018; Buchanan 2020; Miller 2022; Scharre 2023; Ding 2024). This literature, however framed, converges on a pessimistic conclusion for the great majority of

states: meaningful sovereignty, defined this way, requires fabrication capacity, frontier models, and compute infrastructure that no economy outside the United States and China can plausibly assemble. A second and largely separate literature, which we term the soft-sovereignty tradition, locates AI and digital power instead in governance capacity — the ability to shape how technology behaves through law, standards, and market access, independent of who builds the underlying systems (Floridi 2020; Bradford 2020, 2023; O'Hara and Hall 2018). This literature offers a more tractable starting point for states outside the duopoly, but, we argue, an incomplete one when pursued alone: governance unsupported by any material compute capability is vulnerable to exactly the kind of unilateral withdrawal that the Anthropic episode illustrates.

This paper's central conceptual contribution is to integrate these two traditions through the concept of programmable sovereignty, which we define as soft governance anchored by a deliberately minimal degree of hard-stack control — above all pooled compute capacity — sufficient to make governance commitments credible and dependence reversible, without requiring the full-stack autonomy that only the duopoly can currently afford. Programmable sovereignty is neither hard sovereignty (productive autonomy) nor soft sovereignty alone (governance without any material backing); it is a hybrid position. This hybrid position is necessary because rules unsupported by any fallback capacity can be disregarded precisely when they matter most, as the actor who controls the underlying compute, models,

or accelerators retains the unilateral option to withdraw them; a minimal sovereign fallback — sufficient to make exit costly for the withdrawing party rather than catastrophic for the dependent one — converts a hostage relationship into a negotiated and reversible one.

A natural objection follows directly from the N-2 diagnosis in Section 2: if infrastructural dependence is as structurally binding as we have argued, does pooling governance across several non-owning states not simply distribute the same underlying vulnerability across a larger group, rather than resolving it? We accept the premise but feel that the answer is more nuanced. Pooled compute, on the definition above, does not claim to eliminate dependence on the duopoly; it claims only to make that dependence reversible rather than absolute. A coalition that jointly governs even a modest, shared compute base — sufficient to migrate critical workloads, sustain a minimum viable evaluation and deployment capacity, and continue operating during a disruption — faces a different choice than a state with no fallback at all: it can absorb a unilateral withdrawal and negotiate its terms of re-entry, rather than losing access outright with no recourse, as occurred in the Anthropic episode. The claim is therefore one of reversibility, not immunity, and the distinction is precisely what separates programmable sovereignty, on our definition, from the soft-sovereignty literature's governance-only account (cf. Ashraf

and Veneziano 2026, whose critique of “strategic agency” frameworks for AI sovereignty applies most directly to governance claims unaccompanied by any material fallback).

The empirical finding that emerges, detailed in Section 4, is a systematic non-overlap: the economies strongest in programmable-sovereignty governance (the European Union, India, and, on specific levers, Japan and South Korea) are among the weakest in hard-stack production, and the economy with the strongest hard-stack capability (Taiwan, dominant in semiconductor fabrication) is among the weakest in governance capacity. This non-overlap is not incidental to the argument; it is the paper’s central empirical warrant for the institutional response developed in Section 5.

The paper’s second contribution, is to introduce the solution of coordinated specialisation: a coalition architecture,through which complementary N-2 states pool their distinct comparative advantages under a shared governance framework rather than each attempting, separately and unsuccessfully, to replicate the full duopoly stack. We construct two scorecards — a hard-sovereignty scorecard spanning eleven layers of the AI production stack and a programmable-sovereignty scorecard spanning seven governance levers — and apply them to seven major economies (the United States, China, the European Union, Japan, South Korea, Taiwan, and India).We argue that coordinated specialisation is not merely a feasible response but, on the evidence in the two scorecards, the only response available to

N-2 states. Section 6 examines three existing institutional precedents — the European Union's EuroHPC computing initiative, India Stack's digital public infrastructure, and the ASEAN Digital Economy Framework Agreement — each of which demonstrates a necessary component of coordinated specialisation while falling short, individually, of the integrated architecture the paper proposes. Section 7 develops the resulting policy implications, and Section 8 concludes.

The AI Duopoly and the N-2 Problem

2.1 Two strategies of full-stack sovereignty

The United States and China are pursuing observably different but equally exclusionary strategies of AI sovereignty. The United States treats AI infrastructure as an instrument of national strategic competition: the Biden administration's 2025 AI Diffusion Framework sought to restrict the international diffusion of advanced chips and model weights; the Trump administration rescinded the framework on taking office but preserved its underlying geopolitical logic in the subsequent America's AI Action Plan, which emphasises domestic infrastructure build-out and controls targeted at strategic competitors (White House 2025). The cumulative

message of both administrations, despite their policy differences, is that compute, frontier models, and large-scale data infrastructure are instruments of state power rather than purely commercial assets.

China is pursuing a structurally different but equally full-stack strategy: state-directed investment in domestic platform champions, deliberate control over critical upstream inputs such as rare earth elements and refined critical minerals, comprehensive data governance under the 2021 Data Security Law and the Personal Information Protection Law, and explicit political oversight of generative AI applications (Sheehan 2023). The objective is not merely to develop capable AI systems but to ensure that AI development remains aligned with state priorities and insulated from external chokepoints — the same export-control logic that the United States applies to semiconductors, applied by China to critical minerals, talent, and increasingly to AI agents themselves.

Europe, India, and other large economies occupy a structurally different position. The European Union possesses regulatory scale, institutional capacity, and an established legal tradition in technology governance, but depends heavily on non-European cloud infrastructure, frontier foundation models, and semiconductor supply chains for the great majority of its AI deployment. It has rules without infrastructure. Emerging and middle-income economies face a related but distinct version of the same problem: their populations will generate vast quantities of data and their public and private sectors will adopt AI at scale, but without a coordinated

strategy, the economic value generated by that data and that adoption will principally accrue to firms and jurisdictions outside their control. The defining feature of the N-2 condition is therefore not exclusion from the AI economy — most states will be included — but inclusion on terms set unilaterally by one or both members of the duopoly.

2.2 Why AI dependence exceeds monetary dependence

The structural position of N-2 states recalls an established problem in international political economy: dependence on a hegemonic currency under conditions where alternatives are costly to construct and coordination among the dependent states is difficult to achieve (Eichengreen 2011; Cohen 2018). We label the AI-era analogue of this condition the N-2 problem, extending the established N-1 terminology from currency dependence (in which one hegemon's currency serves as the system's reference point) to a setting with two simultaneous hegemons whose interests do not align.

We argue that AI dependence is more consequential than currency dependence for three distinct reasons. First, AI dependence is infrastructural in a stronger sense than monetary dependence. Cross border payments require a reference currency, but this single function does not compare to the breadth of AI's embedding in the

routine operation of public administration, healthcare delivery, logistics, and defence; by the time AI dependence becomes visible to policymakers, it is typically already embedded in code, contracts, cloud architecture, and procurement systems (Farrell and Newman 2019, 2023; Drezner, Farrell and Newman 2021). Second, the costs of switching are categorically higher. A state can redenominate trade contracts in an alternative currency at comparatively low cost; it cannot rebuild, on any comparable timescale, the model, cloud, and semiconductor stack on which its hospitals, courts, and tax administration depend. Third, the window for constructing credible alternatives is closing more rapidly than was the case in the monetary domain, because supply-chain commitments, governance norms, and training-data pipelines are being locked in now, in a period of only a few years, rather than over the multi-decade timescale over which the post-Bretton Woods monetary order consolidated.

Section 4 quantifies the magnitude of this dependence directly. The United States and China dominate global filings of AI-related patents, with the European Union, Japan, and South Korea trailing by a wide margin (WIPO 2024, 2025); the United States retains a decisive lead in benchmarked GPU cluster compute performance, although China is narrowing this gap; China, conversely, dominates the upstream critical-minerals layer on which the entire stack depends, and maintains a

substantially larger and lower-cost installed base of electricity generation capacity, a binding input for AI training and inference at scale (García-Herrero, Grabbe and Källenius 2023; García-Herrero and Mu 2025).

AI Sovereignty: Two Literatures and a Synthesis

3.1 The hard-sovereignty literature: sovereignty as productive autonomy

The dominant scholarly treatment of AI power treats sovereignty as a function of what a state builds. Kai-Fu Lee's AI Superpowers (2018) set the terms of this literature by casting AI development as a two power race between the United States and China, a framing that anticipates closely the duopoly structure underlying our N-2 argument. Ben Buchanan's The AI Triad (2020) provided this race with its first systematic anatomy, decomposing AI capability into compute, algorithms, and data — arguably the conceptual origin of every subsequent AI "stack" framework, though a comparatively narrow one, organised around inputs to a single model rather than the broader institutional architecture that surrounds deployment. Chris Miller's Chip War (2022) narrows the analytical lens further, isolating semiconductor hardware as the single binding constraint on the entire stack and recasting advanced chips as a strategic chokepoint comparable to oil. Paul Scharre's Four Battlegrounds (2023) widens Buchanan's triad by adding talent and institutions as a

fourth domain and arranging the United States, China, and Europe against one another across all four dimensions. Jeffrey Ding's Technology and the Rise of Great Powers (2024) widens the analytical frame still further, arguing that diffusion capacity — a society's capacity to adopt and absorb a general-purpose technology at scale — determines long-run technological power more reliably than the moment of initial invention, which shifts the relevant analytical question from who builds first to who deploys and governs most effectively (Ding 2024). Sastry, Heim, and colleagues' (2024) treatment of compute governance narrows the aperture once more, but toward a different purpose: compute is analytically privileged because it is detectable, excludable, and quantifiable in a way that other inputs are not, which makes the hardware layer not merely an input to be built but a lever that can be governed even by states that do not own it. This last move anticipates the soft-sovereignty literature discussed in Section 3.2, and is the conceptual hinge on which the present paper's synthesis turns. Within this hard-sovereignty literature, the Stanford HAI AI Index (Stanford HAI 2025, 2026) and the Epoch AI datasets supply the empirical infrastructure used in Section 4 to measure national capability in compute, models, investment, and talent.

Taken on its own terms, the hard-sovereignty literature converges on a pessimistic implication for every economy outside the duopoly: sovereignty over AI, defined as productive autonomy, requires fabrication capacity, frontier models, and compute infrastructure that no N-2 state, individually, can realistically assemble.

3.2 The soft-sovereignty literature: sovereignty as governance capacity

A second and largely separate body of scholarship locates AI and digital sovereignty in governance rather than production. Luciano Floridi's (2020) account of the contest for digital sovereignty reframes the relevant question as one of who shapes the digital environment on which a society depends, and argues explicitly for a hybrid model of control in place of full-stack ownership. Anu Bradford's Digital Empires (2023), building on her earlier account of the Brussels Effect (Bradford 2020), demonstrates empirically that the European Union exercises substantial regulatory power not by out-building its technological rivals but through a rights-based regulatory model whose standards diffuse extraterritorially, as global firms find it more efficient to comply with the most stringent applicable jurisdiction than to fragment products by market — regulation, in this account, functions as a form of reach rather than a substitute for production. O'Hara and Hall's (2018) Four Internets makes a structurally parallel argument at the level of underlying values rather than specific rules, identifying four competing governance philosophies — a Silicon Valley open internet, a Brussels rights-based internet, a Washington commercial internet, and a Beijing paternal internet — and showing that the resulting shape of the global digital order is determined by these governance choices as much as by underlying engineering capability. Krasner's (1999) earlier

and more general account of sovereignty as "organised hypocrisy" anticipates this literature's central claim: even classical territorial sovereignty has always combined formal authority with negotiated, frequently informal, control, which supports treating governance capacity as a genuine, if partial, form of sovereignty rather than a residual consolation available only to states that cannot build the underlying stack.

This literature offers N-2 states a more tractable starting point than the hard-sovereignty tradition, because governance capacity, unlike fabrication capacity, does not require matching the absolute scale of duopoly investment. We argue in Section 3.3, however, that it is an incomplete foundation when pursued in isolation from any material capability.

Two further bodies of recent scholarship sharpen this account. Fratini, Hine, Novelli, Roberts and Floridi (2024) provide a systematic comparative review of competing digital-sovereignty models, against which the hard- and soft-sovereignty distinction developed here can be situated; Hummel, Braun, Tretter and Dabrock (2021) supply the standard review of data sovereignty specifically, the layer of the stack on which much of the soft-sovereignty literature's governance claims are built; and Srivastava and Bullock (2024) extend the lineage running from Onuf (1991) through Krasner to contemporary AI governance, reinforcing the claim that sovereignty has long been a contested rather than a settled category even in its classical, territorial form. Most directly relevant to the argument advanced in this paper, however, is Ashraf and

Veneziano's (2026) critical review of "strategic agency" accounts of AI sovereignty — frameworks, structurally similar to the soft-sovereignty literature reviewed here, that locate sovereignty in a state's capacity to act strategically under conditions of irreversible technological interdependence. Ashraf and Veneziano identify three limitations of such frameworks: they underspecify how infrastructural power structurally bounds the strategic choices available to a dependent state; they fail to account for coercion pathways that become visible only under disruption, precisely the dynamic illustrated by the Anthropic episode discussed in Section 1; We take this critique seriously and return to it directly in Section 3.3, where we argue that the minimal hard-stack component built into our definition of programmable sovereignty is intended specifically to answer it, rather than to repeat the vulnerability it identifies in governance-only accounts of sovereignty.

3.3 Programmable sovereignty: a bridging definition

These two literatures rarely engage one another directly: the hard-sovereignty tradition specifies what N-2 states cannot build, while the soft-sovereignty tradition specifies what they might govern, without addressing whether governance alone is

sufficient when the governed systems are owned by an external and potentially adversarial party. This paper's central conceptual contribution is to integrate the two traditions explicitly, through a single, precisely specified concept.

We define programmable sovereignty as soft governance anchored by a deliberately minimal degree of hard-stack control — above all pooled compute capacity — sufficient to make governance commitments credible and dependence reversible. Three elements of this definition require emphasis, because the precision of the concept is important for the remainder of the paper's argument.

First, programmable sovereignty is a hybrid condition, not a point on a single continuous scale running from full dependence to full autonomy. It is categorically distinct both from hard sovereignty (productive autonomy across the stack, approximated only by the duopoly) and from soft sovereignty exercised alone (governance capacity unsupported by any material fallback). A state that scores highly on governance levers but possesses no minimal hard-stack capacity — no pooled compute, no sovereign evaluation infrastructure, no fallback routing option — does not possess programmable sovereignty under this definition; it possesses governance capacity that remains structurally vulnerable to unilateral withdrawal by the party that owns the underlying systems, precisely the vulnerability the June 2026 Anthropic episode demonstrated for the European Union, whose AI Act, Digital

Markets Act, and other soft-sovereignty governance capacity could not prevent two frontier models from being withdrawn from the European market with ninety minutes' notice.

Second, programmable sovereignty does not require, and does not aim at, full-stack autonomy or strategic decoupling from the duopoly. Its logic is to remain efficiently embedded within global AI supply chains — continuing to procure advanced chips, deploy foreign-built foundation models, and exchange data and services across borders, because the efficiency gains from interdependence exceed the costs of autarky in the great majority of cases — while holding a sufficient minimal fallback capacity that the relationship of dependence becomes negotiated and reversible rather than unilateral and hostage-like. The function of this minimal hard-stack component is analogous to a spare wheel: a state that can reach the next point of exit if a current supplier defaults can afford to remain dependent in ordinary circumstances, can negotiate from a credible floor rather than from the edge of a cliff, and can sustain a genuine, mutually beneficial co-dependence with the duopoly rather than a one-sided subordination to it.

Third, and consequently, programmable sovereignty operationalises through a control plane: a defined set of governance and operational mechanisms, detailed in Section 3.4, that allow a state or a coalition of states to direct, constrain, and assure the behaviour of AI systems within their jurisdiction regardless of where the

underlying components were produced or are formally owned. This control plane corresponds directly to the seven levers measured in the programmable-sovereignty scorecard presented in Section 4.

3.4 The control plane: seven mechanisms

Programmable sovereignty operates through seven mechanisms, six of which constitute governance capacity in the soft-sovereignty sense and one of which — pooled compute — supplies the minimal hard-stack substrate that the definition in Section 3.3 requires.

The first mechanism is policy as code: the embedding of compliance requirements, such as data residency rules, audit trails, and safety thresholds, directly into AI systems and public procurement processes, rather than relying exclusively on after-the-fact litigation (Veale and Borgesius 2021; Mariniello 2026). The second is routing control: the capacity to specify which models may process which categories of data, under which jurisdictional conditions, and with which fallback options, so that sensitive workloads in defence, healthcare, taxation, identity management, and the judiciary can be confined to trusted environments while the compute provider functions as an operational intermediary subject to jurisdictional rules (Heim, Anderljung and Belfield 2024). The third is evaluation capacity: the independent

technical ability to test AI systems for safety, security, bias, and reliability, increasingly institutionalised through national AI Safety Institutes operating in several jurisdictions (Anderljung et al. 2023). The fourth is cross-border data compacts: trusted bilateral or multilateral arrangements that permit data to flow across borders without the originating jurisdiction surrendering rights to, or value derived from, that data, exemplified by Japan's Data Free Flow with Trust framework and India's DEPA consent architecture (Nilekani 2015–2024). The fifth is reliability governance: the tracking and constraint of what we term reliability debt, the accumulation of undetected operational risk that arises when AI systems are deployed faster than they can be tested, audited, and corrected.

The sixth mechanism, pooled compute, is the minimal hard-stack substrate specified in the definition above: the joint procurement and shared governance of computing infrastructure, on the model of the European Union's EuroHPC initiative, which both lowers the per-member fiscal cost of maintaining sovereign capacity and creates the possibility of a third compute pole distinct from the hyperscale infrastructure controlled by United States and Chinese firms. The seventh mechanism is market leverage: the use of combined market size to require that global AI providers meet coalition determined standards as a condition of market access. As Bradford (2020, 2023) demonstrates with respect to the Brussels Effect, this mechanism is negligible when exercised by any single N-2 economy but potentially decisive when exercised at coalition scale, and it is this mechanism that

gives the preceding six their practical force: rules that cannot be enforced against a non-compliant global provider are, in practice, recommendations rather than governance.

Together, these seven mechanisms define programmable sovereignty in its most accessible institutional form: governance that is real, legally enforceable, and backed by a minimal but genuine fallback capacity, without requiring full ownership of the underlying stack. We emphasise, consistent with the definition in Section 3.3, that this is the minimum viable form of programmable sovereignty — sufficient to preserve meaningful strategic optionality, but not sufficient, on its own, to eliminate every vulnerability that follows from non-ownership of the full stack. Section 5 develops coordinated specialisation as the mechanism through which N-2 states can move beyond this minimum viable form toward a more robust collective capability.

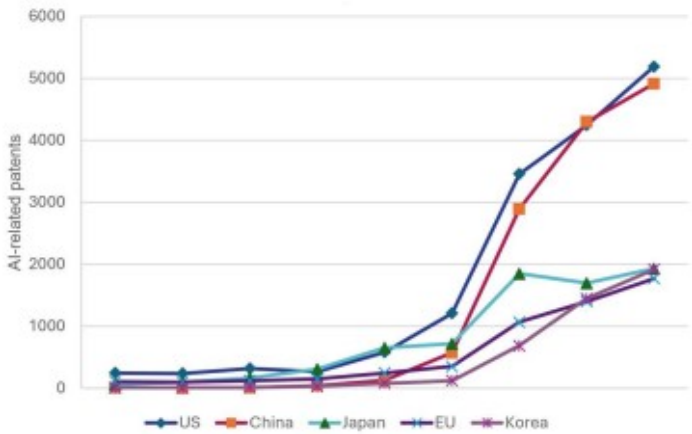
4. Measuring Hard and Programmable Sovereignty

This section provides data to support the conceptual distinction developed in Section 3.

4.1 The Current State of Achievement

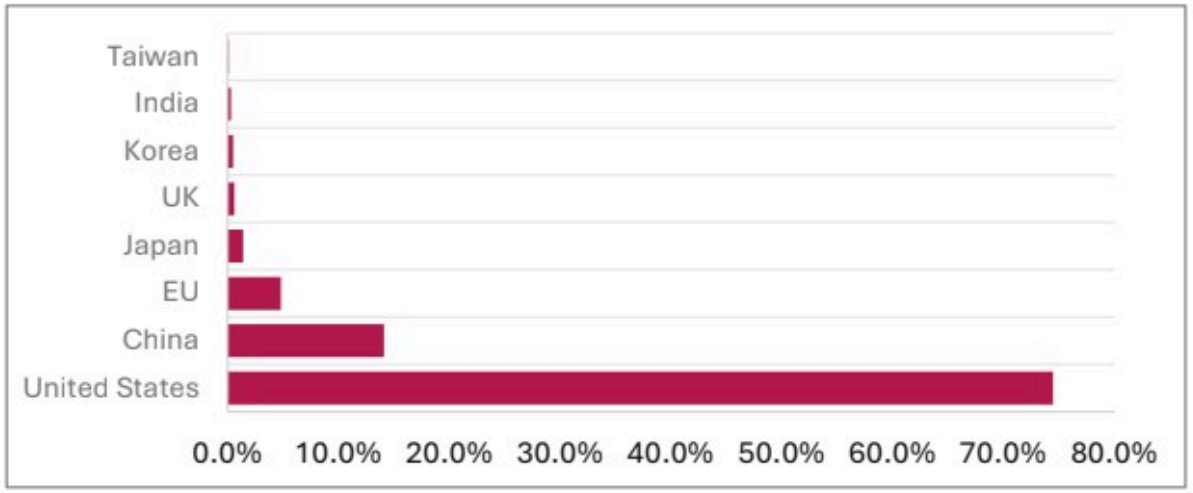
To better gauge how far the rest of the world is from the US or China when it comes to the AI stack, below are some key indicators of where we stand. Firstly, when it comes to innovation, the US and China clearly dominate the number of patents being filed in fields related to AI with Europe, Japan and South Korea trailing at a far distance (Graph 1). Secondly, when it comes to compute, the US clearly dominates (Graph 2) but China is making great strides in this regard. Instead, China dominates the upstream part of the AI stack, such as critical raw materials (Graph 3). In the same way, China has access to more -but also cheaper- energy which is crucial fGraor faster AI adoption (Graph 4).

Graph 1: Number of AI related patents



Source: WIPO

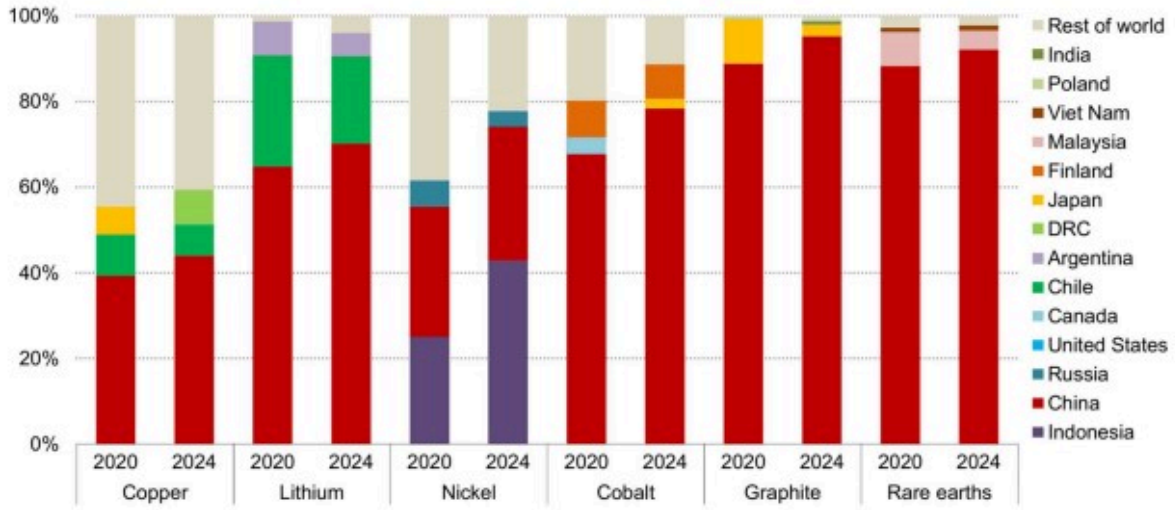
Graph 2: Share of Global AI GPU-Cluster performance (as % of total),



Source X. Data as of May 2026

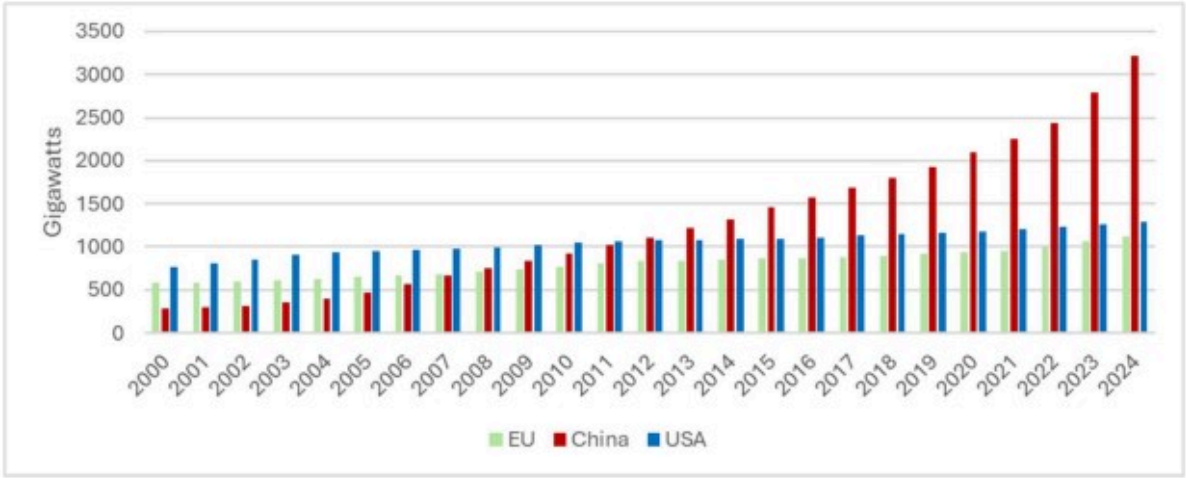
Graph3

Critical mineral processing concentration, 2020 and 2024



Source: Garcia Herrero et at (2023)

Graph 4: Total installed electricity capacity (GW)



Source: García Herrero and Mu (2025)

Table 1 measures hard sovereignty across eleven layers of the AI technology stack; throughout this paper we use hard sovereignty and productive autonomy interchangeably, both denoting the degree to which a state builds, owns, and can reproduce the stack rather than governs it. Table 2 measures programmable sovereignty's constituent governance levers, following the seven-mechanism control plane specified in Section 3.4. Both scorecards apply a four-tier ordinal classification — Leader, Strong, Emerging, Minimal — to seven major economies: the United States, China, the European Union, Japan, South Korea, Taiwan, and India. Classifications are a qualitative synthesis of the primary sources documented

in the footnotes to each table, rather than a single composite index; we report them at this level of precision because the underlying evidence base, while extensive, does not support finer cardinal measurement across so heterogeneous a set of indicators.

Leader Strong Emerging / catching up Minimal

Table 1 — Hard AI sovereignty (who builds and owns the stack)

Layer of the stack	US	China	EU	Japan	Korea	Taiwan	India
Critical materials[1]	Emerging	Leader	Emerging	Emerging	Minimal	Minimal	Emerging
Energy / power[2]	Strong	Leader	Emerging	Minimal	Minimal	Minimal	Emerging
Equipment & lithography[3]	Strong	Emerging	Leader*	Strong	Minimal	Minimal	Minimal
Fabrication (foundry)[4]	Emerging	Emerging	Minimal	Emerging	Strong	Leader	Emerging
Memory / HBM[5]	Strong	Emerging	Minimal	Minimal	Leader	Minimal	Minimal
AI accelerators (design)[6]	Leader	Emerging	Minimal	Minimal	Minimal	Minimal	Minimal
Compute clusters[7]	Leader	Strong	Minimal	Emerging	Minimal	Minimal	Emerging
Software ecosystem[8]	Leader	Emerging	Minimal	Minimal	Minimal	Minimal	Minimal
Foundation models[9]	Leader	Strong	Emerging	Minimal	Minimal	Minimal	Emerging
Private AI investment[10]	Leader	Strong	Emerging	Emerging	Emerging	Minimal	Emerging
Research & talent[11]	Leader	Leader	Strong	Emerging	Emerging	Minimal	Strong

1IEA, Global Critical Minerals Outlook 2025; CSIS, "Beyond Rare Earths: China's Growing Threat to Gallium Supply Chains" (Jul 2025); World Economic Forum / Visual Capitalist refining-share data (2025– 26). China refines ~99% of gallium and ~83% of germanium and leads refined output for 19 of the 20 minerals the IEA tracks.

2U.S. Federal Reserve, "The State of AI Competition in Advanced Economies" (Oct 2025, citing IEA); RBC Global Asset Management (2026); IEEE ComSoc Technology Blog (Feb 2026). China ~3,200 GW installed capacity vs ~1,293 GW (US) and ~1,125 GW (EU); +429 GW added in 2024.

3MarketsandMarkets, Semiconductor Manufacturing Equipment Market (2025); IntelMarketResearch, Semiconductor Process Equipment Outlook 2025–32. ASML holds >95% of EUV lithography; the top five vendors hold 56–66% of all equipment revenue.

4TrendForce foundry rankings 2025, via Statista, Visual Capitalist and The Motley Fool. TSMC ~70% share, Samsung ~7%, SMIC ~5%, GlobalFoundries ~4%. 5TrendForce and general industry reporting (2025). SK Hynix and Samsung (Korea) lead high-bandwidth memory; Micron (US) is third.

6Brookings, "Competing AI strategies for the US and China" (Apr 2026); MERICS, China's drive toward AI self-reliance; Reuters, CFR and IFP chip-gap analyses (Dec 2025). Nvidia, AMD and Google lead; Huawei Ascend is ~one generation behind.

7Epoch AI, AI supercomputers dataset (2025); Stanford HAI, AI Index 2025. The US hosts ~75% of benchmarked GPU-cluster performance vs China's ~14%, and holds ~9× China's AI compute. 8MERICS, China's drive toward AI self-reliance; industry analysis. Nvidia's CUDA remains the dominant developer moat; Huawei's CANN is China's main alternative. 9Stanford HAI, AI Index 2025. The US produced 40 notable models in 2024 vs China's 15 and Europe's three, with Chinese models reaching near benchmark parity.

10Stanford HAI, AI Index 2025. US private AI investment reached $109.1B in 2024 vs China's $9.3B and the UK's $4.5B.

11Stanford HAI, AI Index 2025. China leads in publication and patent volume; the US leads in highly-cited research and top-talent concentration.

Table 2 — Programmable sovereignty (who can govern AI without building it)

Lever (control plane)	US	China	EU	Japan	Korea	Taiwan	India
Build rules into the system[12]	Emerging	Strong	**Leader**	Emerging	Strong	Emerging	Strong
Control where data & decisions live[13]	Strong	**Leader**	**Leader**	Emerging	Emerging	Minimal	Strong
Check what vendors claim[14]	Strong	Strong	**Leader**	Emerging	Emerging	Minimal	Emerging
Share data without losing control[15]	Emerging	Emerging	**Leader**	Strong	Emerging	Minimal	**Leader**
Stop hidden risk piling up[16]	Emerging	Strong	**Leader**	Emerging	Emerging	Minimal	Emerging
Pool computing as a shared utility[17]	Strong	Strong	Emerging	Emerging	Strong	Emerging	Emerging
Use the size of your market[18]	**Leader**	Strong	**Leader**	Emerging	Emerging	Minimal	Strong

12EU AI Act and GPAI Code of Practice (European Commission; Jones Day, 2025); India DPDP Act & Rules 2025 with the DEPA consent rails (MeitY; Asia Society, 2025); China Generative AI Services Management Measures (OneTrust; GDPRLocal); Korea AI Basic Act (IAPP).

13EU Data Act (applicable 12 Sep 2025) and sovereign-cloud requirements (Alice Labs, EU AI Infrastructure report 2026); China data-localization under PIPL and the Data Security Law; India DPDP data residency rules.

14EU AI Office and Testing & Experimentation Facilities (European Commission); the US, Korean and Japanese AI Safety Institutes; China's model registration and security review. 15EU Data Governance Act and Common European Data Spaces; India's Account Aggregator / DEPA framework (Asia Society, 2025); Japan's Data Free Flow with Trust, launched at the 2019 G20 Osaka summit (Japan Digital Agency).

16EU AI Act high-risk obligations with DORA and NIS2; India DPDP breach reporting and the Data Protection Board (S.S. Rana & Co., 2026); Korea AI Basic Act high-impact risk-management duties. 17EU EuroHPC AI Factories and InvestAI gigafactories (EuroHPC JU, Oct 2025); India's IndiaAI Mission shared GPUs (34,000 now, 100,000 targeted by end-2026); Korea's sovereign-cloud and GPU build out.

18GDPR, the Digital Markets Act and Digital Services Act, with EU AI Act fines up to €35M or 7% of global turnover; India Stack's population-scale rails (UPI, Aadhaar) disciplining private firms.

Tables 1 and 2, read together show that hard-stack production capacity and programmable-sovereignty governance capacity are distributed in a complementary fashion across the seven economies examined.

The United States and China score highly across most rows of Table 2, but, consistent with the definition in Section 3.3, their position does not constitute programmable sovereignty as we define it: their governance capacity derives from owning the stack outright (the United States) or directing it comprehensively through the state (China), not from governing a stack they do not own. Both economies, in other words, possess hard sovereignty; Table 2's scores for these two cases should be read as a description of how full-stack owners exercise comprehensive control, not as evidence that they have constructed a programmable-sovereignty solution to a dependence problem they do not face.

The European Union presents the clearest instance of programmable sovereignty's target condition and its central limitation. The EU records Leader status on five of seven levers in Table 2, grounded in verifiable institutional fact: the AI Act (Regulation 2024/1689) embeds conformity assessment and post-market monitoring by design into high-risk systems; the EU AI Office, established in February 2024, provides institutionalised evaluation capacity for general-purpose models; NIS2 (2022) imposes resilience requirements on critical digital infrastructure; and near-universal global compliance with the GDPR provides the clearest existing evidence of Brussels-Effect market leverage in the digital domain (Bradford 2020). Yet the EU records Minimal or Emerging status on nine of eleven layers in Table 1, with its sole Leader position — equipment and lithography — resting almost entirely on a single firm, ASML. This combination is precisely what our definition predicts will be vulnerable: governance leadership unsupported by a sufficient hard-stack fallback. The clearest empirical illustration is the June 2026 Anthropic episode itself. The European Union's extensive regulatory apparatus could not prevent two frontier models from being withdrawn from the European market with ninety minutes' notice, because that apparatus governs systems the EU does not own and, critically, had not yet been paired with the pooled-compute fallback that Section 3.3 identifies as constitutive of programmable sovereignty proper. The

EU at present therefore exercises soft sovereignty extensively but programmable sovereignty only partially, pending the build-out of its pooled-compute capacity under EuroHPC and the AI Factories initiative (examined in Section 6).

India occupies a comparable but smaller-scale position: broadly Emerging-to-Minimal across the hard stack layers in Table 1, but a Leader on data compacts (via the DEPA consent architecture and Account Aggregator framework) and Strong across several other governance leverss. Japan and South Korea occupy more specialised niches within the governance scorecard: Japan as the principal architect of the Data Free Flow with Trust framework, agreed at the 2019 G20 Osaka summit, though its domestic AI Promotion Act remains deliberately soft law with no associated financial penalties; South Korea through its AI Basic Act, which took effect in January 2026 as the world's second comprehensive national AI statute with explicit extraterritorial reach, paired with a substantial sovereign-cloud investment programme. Taiwan presents the empirical mirror image of the European Union: a Leader position on a single, globally indispensable hard-stack layer (semiconductor fabrication) combined with Minimal status across every governance lever in Table 2.

The substantive implication of this non-overlap is that no single N-2 economy examined here possesses, on its own, the combination of governance capacity and minimal hard-stack fallback that the definition of programmable sovereignty requires. The European Union, India, Japan, and South Korea collectively possess

the governance capacity; Taiwan and South Korea collectively possess critical fabrication and memory capacity that none of the governance leaders possesses individually. This is not a coincidental empirical pattern. It is the structural condition that coordinated specialisation, as an institutional design, is built to exploit.

5. Coordinated Specialisation: From Individual Insufficiency to Coalition Sufficiency

Section 4's empirical finding — that governance capacity and hard-stack production capacity are concentrated in different, non-overlapping sets of economies — provides the support for this paper's core proposal. If no single N-2 economy can independently satisfy the definition of programmable sovereignty developed in Section 3.3, the relevant policy question is whether a coalition of complementary economies can satisfy it collectively. We argue that it can, through a coordinated specialisation architecture in which each participating state contributes its comparative advantage under a shared governance framework, rather than each state attempting, in parallel and at much greater aggregate cost, to replicate the capabilities the others already possess.

The intellectual lineage of this argument runs through the industrial-policy literature on coordinated public action in strategic sectors characterised by high fixed costs, network effects, and first-mover advantages that markets alone do not efficiently overcome (Mazzucato 2013; Rodrik 2022; Aiginger and Rodrik 2020; Cherif and Hasanov 2019). The most directly relevant historical precedent is the European aerospace consortium Airbus. Airbus did not emerge from any single European state's attempt to construct a full national aerospace industry capable of matching Boeing; it emerged from explicit national specialisation — French-built airframes, British-built wings, German-built fuselage sections, Spanish-built tail assemblies — coordinated under a shared governance structure and underwritten by a combined market scale that no single participating state could have provided alone (García-Herrero and Martens 2026). The Airbus precedent demonstrates three design features that we argue transfer directly to the AI domain: deep functional specialisation rather than duplicated effort, governance arrangements agreed prior to, and conditioning, joint investment, and patient public capital sustained over a multi-decade horizon rather than calibrated to short-run political cycles.

We note, without resolving the question here, that coordinated industrial strategies of this kind have a mixed empirical record, and that the literature on industrial policy identifies systematic failure modes — binding state-capacity constraints, weak enforcement of conditionality on participating firms or states, and rent-seeking among coalition members — that are not unique to the AI domain

(Bulfone 2023; Juhász and Lane 2024). The Airbus case is, on the available comparative evidence, a successful rather than a representative outcome among coordinated multi-state industrial ventures, and the coalition design proposed below should accordingly be read as institutionally feasible rather than institutionally guaranteed; Section 7.4's emphasis on sequencing governance before infrastructure is intended specifically to mitigate common causes of coordinated industrial projects underperforming their design.

Applied to the present case, the coordinated-specialisation architecture aggregates the European Union's governance leadership (Table 2) with the complementary hard-stack specialisations of Taiwan (fabrication), South Korea (memory and, increasingly, sovereign compute), Japan (data-governance architecture), and India (population-scale digital public infrastructure and a rapidly expanding sovereign compute base), with sovereign capital as a plausible source of the patient, multi-decade financing that the Airbus precedent indicates is necessary. No single one of these economies, our scorecards demonstrate, could independently satisfy the definition of programmable sovereignty developed in Section 3. Pooled under a shared governance architecture, however, the coalition collectively possesses both the governance levers and the minimal hard-stack fallback — above all, jointly governed compute capacity — that the definition requires, at a scale that no individual member, and notably neither the United States nor China acting unilaterally, can straightforwardly deny it.

The coordinated-specialisation architecture proposed above raises a further question that the comparative-advantage logic alone does not resolve: what sustains coalition membership once formed, and what prevents defection once an individual member's own capability matures? A state that builds sufficient domestic compute capacity, for instance, faces an incentive to withhold it from coalition wide pooling once its own minimal fallback is secure, free-riding on the market leverage that the coalition's combined scale continues to provide. We do not offer a full game-theoretic treatment of coalition stability here; this is properly a question for the collective-action and alliance literatures (cf. the conditionality-enforcement problem identified in Bulfone, Ergen and Maggor 2024), and we flag it explicitly as an open question for future work rather than treating coalition formation as self sustaining once comparative advantages are identified. The governance-before-infrastructure sequencing recommended in Section 7.4, and the shared-metrics proposal in Section 7.5, are partial institutional responses to this defection risk — binding commitments and observable compliance are the standard tools for sustaining cooperation under incentives to free-ride — but we do not claim they are sufficient on their own.

6. Existing Institutional Precedents and Their Limits

This section examines three existing institutional arrangements that each demonstrate a necessary component of the coordinated-specialisation architecture proposed in Section 5, while individually falling short of the integrated design the paper proposes.

6.1 EuroHPC: pooled infrastructure under shared governance

EuroHPC constitutes the clearest existing proof of concept for the minimal hard-stack substrate that programmable sovereignty, on our definition, requires. The EuroHPC Joint Undertaking pools European Union, member-state, and private resources to procure and operate world-class computing systems — including LUMI (Finland), Leonardo (Italy), and JUPITER (Germany, Europe's first exascale system) — under jointly negotiated access rules rather than national ownership.[1] Building on this base, the European Commission's AI Factories initiative, announced as part of the 2024 AI Innovation Package, federates and upgrades these systems into a network of AI-optimised hubs offering compute access to start-ups, small and medium enterprises, and researchers.[2] A larger and more frontier-capable tier, the AI Gigafactories, is financed through the €20 billion InvestAI facility, but remained at the call-for-expressions stage as of early 2026, with the EuroHPC mandate formally amended to incorporate gigafactories only in January 2026. EuroHPC therefore

establishes that infrastructure pooling under shared governance is operationally feasible at continental scale; its frontier-capable tier, however, remains prospective rather than operational, and its governance has not yet been extended to the full set of control-plane mechanisms — evaluation, routing, and reliability governance — that programmable sovereignty, on our definition, presupposes.

6.2 India Stack: programmable architecture and its limits

India Stack — comprising Aadhaar, the Unified Payments Interface, the Account Aggregator framework, and the Open Network for Digital Commerce — demonstrates that population-scale digital public infrastructure designed and governed from the Global South can operate at a level of technical sophistication and reliability comparable to any equivalent system globally. Its architectural significance for the present argument is that public-purpose digital rails, on which private firms, financial institutions, and public agencies subsequently build, can be constructed without requiring state monopoly over the underlying digital services. Public priorities — identity inclusion, low-cost payment access, consent-based data sharing, and cross-platform interoperability — are encoded directly into the architecture rather than imposed externally through subsequent regulation. India

Stack is accordingly a powerful institutional precedent for the kind of programmable architecture that coordinated specialisation requires, and its Leader classification on data compacts in Table 2 reflects this operational reality directly.

The translation from digital public infrastructure to AI sovereignty proves harder. The IndiaAI Mission, targeting 100,000 GPUs of sovereign compute capacity by the end of 2026, with approximately 34,000 already deployed, reveals simultaneously what is achievable and what remains structurally difficult. India possesses substantial talent — the highest AI skill-penetration rate globally, at 2.8 times the global average — alongside proven digital-infrastructure design capability and a domestic market of considerable scale, but remains dependent on foreign-manufactured chips, foreign cloud infrastructure, and foreign-built foundation models for the most capable tier of AI systems. The lesson, for the present argument, is that AI sovereignty is structurally more demanding than digital services sovereignty: India Stack demonstrates what programmable architecture looks like when correctly designed, while the IndiaAI Mission demonstrates why a coalition-level strategy, rather than a purely national one, is required at the AI layer specifically.

6.3 The ASEAN Digital Economy Framework: regional governance as a partial building block

The ASEAN Digital Economy Framework Agreement, under negotiation through 2025–2026, aims to establish shared rules governing digital trade, cross-border data flows, and AI deployment across Southeast Asia, combining Singapore's regulatory sophistication, Indonesia's demographic scale, Vietnam's manufacturing base, and Malaysia's expanding data-centre capacity. The agreement stops well short of the operational mechanisms that coordinated specialisation, on the architecture proposed in Section 5, requires: it does not pool compute capacity, does not establish joint model evaluation, and does not impose collective procurement requirements at coalition scale. As an institutional building block for a more comprehensive future partnership, the agreement is significant; as a sovereignty architecture in its own right, it remains incomplete.

Read together, these three precedents demonstrate both the institutional feasibility of coordinated specialisation and the scale of the integration gap that remains. EuroHPC demonstrates pooled infrastructure; India Stack demonstrates programmable architecture; the ASEAN framework demonstrates regional governance coordination. None of the three, individually, demonstrates the integrated combination — governance, evaluation, routing, and pooled compute operating under a single shared framework across a coalition spanning multiple regions — that the architecture proposed in Section 5 requires.

7. Policy Implications

The analysis developed in Sections 3 through 6 supports a change in strategic posture rather than a discrete catalogue of measures. Its central implication is that no individual actor in the N-2 world, the European Union included, can secure programmable sovereignty alone, and that attempting to do so by reconstructing the full hard stack is both prohibitively costly and, on the evidence of Section 4, unnecessary, since the constituent capabilities already exist across the prospective coalition, only in dispersed form. The first-order policy task, accordingly, is to convert these dispersed and individually insufficient capabilities into a pooled and jointly governed whole. We develop this implication along five dimensions.

7.1 Lead with governance, not production

The control plane developed in Section 3.4 — policy as code, routing control, evaluation capacity, data compacts, reliability governance, pooled compute, and market leverage — is the domain in which N-2 states retain genuine and immediately exercisable agency. Policy should accordingly prioritise building and harmonising these levers over subsidising full-stack production efforts that, on the evidence assembled here, cannot succeed at any plausible budget. For the

European Union specifically, this implies deploying its existing regulatory leadership as coalition-scale leverage rather than continuing to exercise it in isolation: the Brussels Effect compounds in force when exercised through combined market and procurement weight across a coalition, and weakens when the EU regulates alone against systems it cannot itself replace.

7.2 Anchor governance with a minimal hard fallback

The June 2026 Anthropic episode demonstrated directly that governance commitments unsupported by any material capacity can be nullified within minutes. The implication, consistent with the definition developed in Section 3.3, is not that the coalition should attempt to construct a rival full stack, but that it should secure sufficient shared, jointly governed capacity — above all pooled compute, and assured access to a small number of genuinely irreplaceable chokepoint assets — to render its governance commitments credible and its dependence reversible. The objective, precisely specified, is strategic optionality: the demonstrated ability to route critical workloads to trusted alternative infrastructure and to negotiate from a credible floor rather than from crisis, while remaining deliberately and efficiently embedded within global AI supply chains rather than pursuing autarky.

7.3 Organise around demonstrated comparative advantage

The Airbus precedent is instructive precisely because no individual participant attempted to construct the complete aircraft independently. The European Union's regulatory and lithography-equipment strengths, Taiwan and South Korea's fabrication and memory leadership, Japan's data-governance architecture, India's population-scale digital public infrastructure, and sovereign capital are complementary capabilities rather than competing claims to the same function. Each prospective coalition member should deepen and make available, under shared rules, its area of demonstrated comparative advantage rather than dissipating scarce resources across stack layers in which it has no realistic prospect of leadership. Coordinated specialisation, not parallel duplication, is the mechanism that converts individually insufficient capabilities into a collectively sufficient stack.

7.4 Sequence governance before infrastructure

Specialisation generates programmable sovereignty only when it is conducted under governance terms agreed in advance — covering data residency, evaluation and audit protocols, intellectual property and benefit-sharing arrangements, reliability obligations, and dispute resolution. Absent this sequencing, pooling capability simply relocates dependence rather than resolving it: a state that supplies compute capacity under another coalition member's unilateral governance terms has exchanged one external master for another. Both the Airbus and EuroHPC precedents indicate that a shared governance framework, open to associated partners from the outset, should precede and condition any joint infrastructure investment. The founding act of the proposed partnership should accordingly be an international agreement establishing this governance framework, not a construction project.

7.5 Make programmable sovereignty empirically measurable

The N-2 problem is dangerous in part because dependence accumulates without becoming salient to policymakers until it is activated, embedded in code, contracts, and procurement decisions made well before the dependence becomes visible. The coalition should accordingly commit to a small set of shared, comparably defined indicators — for example, the speed of trustworthy AI deployment within regulated

public-sector applications, the share of critical workloads operating without completed independent evaluation or any sovereign fallback option, and the share of critical workloads for which no coalition-internal alternative currently exists. Shared metrics of this kind convert programmable sovereignty from an aspirational posture into an empirically observable condition and provide member states with an evidentiary basis for assessing whether their collective autonomy is increasing or eroding over time — a research agenda for future quantitative work that the present paper's qualitative scorecards are designed to anticipate rather than to substitute for.

These five policy directions are deliberately specified as a strategic posture rather than a fixed institutional design. The specific instruments through which the partnership is realised — which body convenes it, how compute access is reciprocated among members, how evaluation capacity is pooled, how patient capital is mobilised — are properly matters for negotiation among the governments concerned, and multiple institutional designs could satisfy the underlying logic this paper establishes. That logic is threefold: that programmable sovereignty, as defined in Section 3.3, is attainable for N-2 states, but only collectively rather than individually; that it requires soft governance anchored by a minimal hard-stack fallback, not soft governance alone; and that it must be constructed within a closing window, as supply chains, governance norms, and data-infrastructure commitments are presently being locked in. The choice facing the European Union and its

prospective coalition partners is consequently not between dependence and autarky, but between dependence that is collectively managed and dependence that is unilaterally imposed — and programmable sovereignty, pursued through coordinated specialisation, is the institutionally realistic route to the former.

8. Conclusion

This paper has argued that the contemporary organisation of artificial intelligence production and governance is generating an N-2 world, in which only the United States and China can credibly pursue full-stack technological sovereignty, and every other state faces a deepening dependence on one or both. For N-2 states, confronting this condition by attempting to reconstruct the full stack is neither financially affordable nor, on the evidence assembled in this paper, achievable within any policy relevant timeframe.

The alternative we have developed, programmable sovereignty, rests on a definition we have held fixed throughout: soft governance anchored by a deliberately minimal degree of hard-stack control — above all pooled compute capacity — sufficient to make governance commitments credible and dependence reversible. This is neither hard sovereignty (productive autonomy, available only to the duopoly) nor soft sovereignty pursued alone (governance unsupported by any material fallback, and consequently vulnerable to exactly the kind of unilateral withdrawal documented in

this paper's opening examples). It is a hybrid condition, and its hybrid character is the paper's principal conceptual contribution to the literatures reviewed in Section 3.

The two scorecards developed in Section 4 demonstrate that the capabilities this hybrid condition requires already exist, in aggregate, across a plausible coalition of N-2 economies, but in dispersed and individually insufficient form: the economies strongest in governance capacity are, with notable consistency, weakest in hard-stack production capacity, and vice versa. This systematic non-overlap is the empirical foundation for coordinated specialisation — pooling the European Union's regulatory leadership, Taiwan and South Korea's fabrication and memory strength, Japan's data-governance architecture, India's digital public infrastructure, and sovereign capital, under a shared governance framework modelled on the European aerospace consortium Airbus. No single one of these economies could independently assemble a credible programmable-sovereignty control plane; collectively, our analysis indicates that they could.

Three limitations of the present analysis are worth stating explicitly. First, the scorecards in Section 4 measure governance and production capacity but not the conditions under which a coalition forms and persists; we have flagged this collective-action question in Section 5 as a matter for future, more formally modelled work rather than resolved it here. Second, the analysis is throughout a capacity based account of sovereignty — what a state or coalition can do — and

does not address the distinct normative question of whether a given exercise of governance capacity, by a coalition or by any of its individual members, is legitimate in the eyes of the populations whose data and behaviour it governs (cf. the normative turn in the digital-sovereignty literature, e.g. Fossa 2024). We treat this as an important and currently unaddressed extension of the framework rather than as a question this paper resolves. Third, the seven-economy scope of our scorecards, justified on data-comparability grounds in Section 4, means the empirical claims in this paper should be read as applying most directly to that set, with the broader coalition discussed in Section 7 extended by qualitative argument rather than by the same level of scorecard evidence.

The institutional window for accomplishing this is open but, on the evidence reviewed in Sections 1 and 2, closing. The June 2026 directive restricting international access to two frontier AI models demonstrated that AI dependence can be activated within minutes, without advance warning and without practical recourse for the affected party. Supply-chain commitments are being established years in advance of their operational deployment; governance norms are being set in forums from which most affected states remain absent; and data-infrastructure pipelines are being constructed in ways that will encode particular values and priorities for decades to come. Sovereignty in the era of artificial intelligence will not be secured solely through constitutions and international treaties; it will be secured, or not, through architectures, governance code, and the institutional

arrangements that determine how AI systems actually operate. For the great majority of the world's states, the European Union included, that sovereignty will be constructed collectively, through coalitions of the kind this paper has specified, or it will not be constructed at all.

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Appendix: Coalition Feasibility — Supporting Data

This appendix reports supporting quantitative data referenced in Section 7 on the financial feasibility of coalition-scale coordinated specialisation.

Table A1. Principal coalition sovereign wealth funds (2025)

Fund	Country	AUM 2025 (US$B)	Source

Government Pension Fund Global	Norway	~2,200	Multiple confirmed (end-2025)
Abu Dhabi Investment Authority	UAE	~1,110	SWF Institute 2025
Public Investment Fund	Saudi Arabia	~1,000–1,150	SWF Institute 2025
GIC Private Limited	Singapore	~936	GIC Annual Report, Aug. 2025
Temasek	Singapore	~320	Temasek Annual Report 2025
Four-fund total		**~5,246–5,396**	
2% mandate, annual		**~105–108**	**Comparable to current EU+China private AI investment combined**

Table A2. Cumulative AI investment by selected coalition members to 2024

Country	Cumulative AI investment, 2013–2024 (US$B)	Estimated 2025 annual (US$B)	Source
India	11.1	~3.5	Stanford AI Index 2025
Japan	5.9	~1.5	Stanford AI Index 2025
South Korea	8.96	~1.0	Stanford AI Index 2025
Australia	3.88	~0.8	Stanford AI Index 2025

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